

Qatar-1b

R-0624

Benchmark run · workflow W8-benchmark-deep · record deep-bench_Qatar-1_240628 · created 2026-07-25T13:15:45+00:00

BENCHMARK: NON-DETECTION

Results

Mid-Transit Time (Tmid)	2460489.9131 +/- 0.0029 BJD_TDB
Ratio of Planet to Stellar Radius (Rp/R*)	0.191 +/- 0.055
Transit depth (Rp/Rs)^2	3.66 +/- 2.11 [%]
Orbital Inclination (inc)	88.67 +/- 2.84
Airmass coefficient 1 (a1)	22.23 +/- 16.11
Airmass coefficient 2 (a2)	-2.74 +/- 1.7
Scatter in the residuals of the lightcurve fit is	66.14 %
Best Comparison Star	#2 - [452, 128]
Optimal Aperture	0.75
Optimal Annulus	3.0
Transit Duration (day)	0.0824 +/- 0.0084

Accuracy vs published ephemeris (ExoClock/NEA)

Rp/R $\blacksquare$ fit vs published	0.191 $\pm$ 0.055 vs 0.1463 (deviation 0.48 σ)
Duration fit vs published	0.0824 d vs 0.0692 d (deviation 0.92 σ)
Mid-time O-C	-4.7 min
Depth SNR	2.0
Residual scatter	66.14 %

Light curve

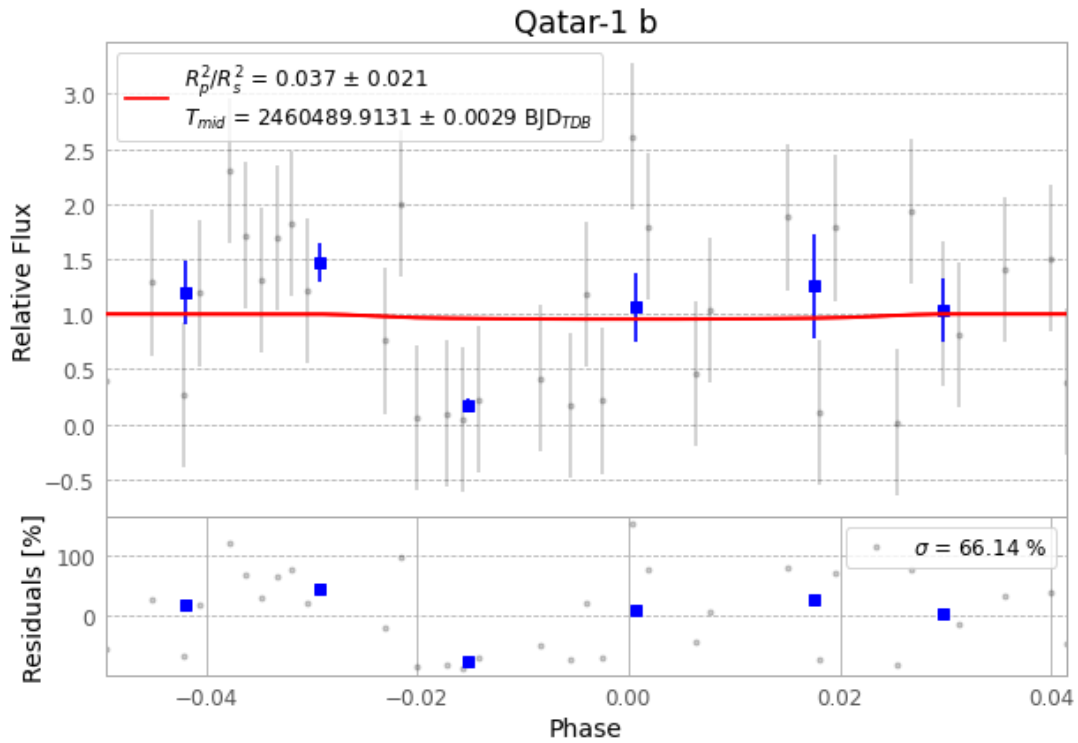


Figure 1 — FinalLightCurve_Qatar-1 b_2024-06-28.png (SHA-256 06ac9341d8a66142...)

Method & software

EXOTIC 4.3.1 aperture photometry with nested-sampling transit fit (ultranest 3.6.5); priors from the NASA Exoplanet Archive. Camera CCD, binning 1x1, filter CV, target pixel [438, 181], 3 comparison star(s). Exact configuration: parameters.inits.json in record.json (SHA-256 2f755857cddf0df8...).

exotic 4.3.1 · astropy 6.1.7 · photutils 2.0.2 · numpy 1.26.4 · scipy 1.14.1 · twirl 0.5.1 · astroquery 0.4.11 · ultranest 3.6.5 · python 3.12.13 · pipeline_commit b365494

Data provenance

64 FITS frames (42 MB), Qatar-1240628080916.FITS → Qatar-1240628111815.FITS. Every input frame is SHA-256-fingerprinted in record.json; raw frames available on request and verifiable against that manifest. Artifacts: bench-Qatar-1-240628.log (77e82349b95c...), FinalLightCurve_Qatar-1 b_2024-06-28.png (06ac9341d8a6...), FinalLightCurve_Qatar-1 b_2024-06-28.csv (20b5a8c2db6f...)

Compute resources

Wall time 175.4 s · peak memory None MB · host mac-mini-m1-8gb (macOS-26.4-arm64-arm-64bit)

Observer: — · AAVSO obscode ABGA · Automated pipeline with human review — nothing is submitted without a human approving this specific result. Data license CC-BY-4.0. Generated 2026-07-25 13:15Z.